

Handwritten Digit Recognition using Convolutional Neural Networks on MNIST

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1. Introduction

Optical character recognition and automated document processing remain foundational challenges in computer vision and machine learning applications across industry and academia. Accurately classifying handwritten numeric digits is essential for automated postal mail sorting, commercial bank cheque processing, tax document ingestion, and digital archival workflows. This proposal implements a standard supervised deep learning classifier to recognize isolated grayscale digits ranging from zero to nine with high throughput.

2. Methodology

We evaluate a sequential Convolutional Neural Network architecture built using standard 3x3 convolutional filters, batch normalization layers, ReLU activation functions, and max-pooling layers for spatial subsampling. The model is trained on the canonical MNIST benchmark dataset comprising 60,000 training images and 10,000 testing images of dimension 28x28 pixels with centered numerals. Optimization is conducted using categorical cross-entropy loss and the standard Adam optimizer initialized with default learning parameters. Dropout regularization (rate 0.25) is applied before the final softmax dense classification layer to prevent parameter overfitting.

3. Results

The convolutional model converges within fifteen training epochs, achieving a test accuracy of 98.7% on the MNIST evaluation partition. Confusion matrix diagnostics indicate that residual classification errors predominantly occur between visually ambiguous numeral pairs such as digits four and nine, or three and five.

4. Conclusion

The study demonstrates that standard convolutional networks reliably achieve high accuracy on benchmark digit recognition datasets. However, because the MNIST benchmark is widely considered saturated and well-established in standard classroom tutorials, future investigations should evaluate transferability across complex real-world alphanumeric datasets.